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Estimating Career Benefits from Online Community Leadership: Evidence from Stack Exchange Moderators

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Abstract. Many IT professionals seek to improve their job prospects by engaging as leaders of online communities, for example, by serving as a moderator or admin. We investigate whether such community leadership leads to (causal) improvements in individuals' careers. We assemble a data set, including job histories of IT professionals who have sought election as moderators (mods) in Stack Exchange question-and-answer communities, between 2010 and 2020. We estimate the career benefits of moderatorship under two complementary identification strategies: difference-in-differences (DID) and regression discontinuity (RD). We observe qualitatively consistent results under each design, finding that election to a moderator role has a significant, causal, positive effect on job mobility. We estimate that moderatorship increases the probability of a job change by between 4.7 and 12.3 percentage points over the two years following the election. We also report a series of secondary analyses that speak to associated salary increases and show evidence consistent with the notion that social capital and signaling play a role. Our findings help us understand the benefits of online community leadership, and they extend our understanding of the motivations for online community engagement.

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Keywords: online community leadership • career benefits • regression discontinuity • difference-in-differences • social capital

1. Introduction

A growing body of evidence indicates that IT professionals' contributions toward online communities are motivated by career concerns (Lerner and Tirole 2002, Wasko and Faraj 2005, Roberts et al. 2006, Xu et al. 2020).¹ Among such contributions, perhaps the most significant are those made by individuals who take on a formal leadership role, for example, serving as community moderators or managing open-source software projects (Faraj et al. 2015, Johnson et al. 2015, Safadi et al. 2021). Serving in a leadership capacity can require several hours of work per day, and the work is typically not compensated, implying opportunity costs. Despite the claim that community contributions can provide career benefits, little evidence exists to support it, a gap we seek to address here.

Identifying the career benefits of community leadership is difficult because detailed data on individuals' careers are generally difficult to obtain. Community leaders seldom share information publicly with the community about their backgrounds or personal lives.

When such information is disclosed, it tends to reflect points in time prior to—rather than following—their tenure in a community leadership position, because individuals often disclose such information when seeking to build a rapport with the community they hope to lead. Even if full job trajectories were available, it is difficult to determine what portion of career advancements truly derived from individuals' assumption of community leadership roles. The decision to engage in a community is endogenous, and thus a comparison of leaders' career trajectories against those of non-leaders would very likely embed noncausal sources of association; that is, differences between leaders and non-leaders would reflect confounding factors that simultaneously drive both community engagement and career progression (e.g., talent, ability, grit). It is therefore not surprising that evidence on the career benefits of community leadership is scarce and, at best, correlational.

We seek to overcome these challenges via a novel integration of alternative data sources and complementary empirical strategies. We study Stack Exchange, a

platform that hosts question-and-answer communities on a variety of subjects (e.g., software development, sciences, or law). Each community is led by a few individuals, so-called moderators. A moderator's job involves adding or removing members, reviewing and rejecting posts, and resolving conflicts. Importantly, moderators are democratically elected by each community, which allows us to assess the job outcomes for any given candidate as a function of whether they won (versus lost) an election. Our sample is thus comprised of candidates who ran for election in several Stack Exchange communities between 2010 (when elections first began) and 2020.

We manually identified candidates' LinkedIn profiles and collected their posted job histories. To operationalize career benefits, we focus on the occurrence of job changes. Job changes are a meaningful indicator of career benefits because job changes are most often associated with career advancement (Topel and Ward 1992, Zippia 2023), and most job changes are voluntary in nature, implying that they are more likely to result from worker preferences than layoffs or firings (BLS 2018).²

We rely on two complementary empirical strategies to identify causal estimates. Our first strategy is a difference-in-differences (DID) design with matching, wherein we compare the job changes before and after elections, contrasting those outcomes between moderatorship candidates successfully elected and similar yet nonetheless unelected candidates. This DID yields a global average treatment effect (ATE) but relies on a strong set of assumptions regarding omitted variable bias. Our second identification strategy is thus a regression-discontinuity design (RDD). The RDD requires relatively weaker assumptions to establish causality (Imbens and Lemieux 2008) but restricts the candidate field to a smaller sample and yields only a local average treatment effect (LATE). We embed both estimations within a hazard modeling framework, consistent with other recent work studying factors that influence career shifts (see, e.g., Card et al. 2007, Chetty 2008, and Yang 2017).

We observe consistent results under both estimation strategies; we find that taking on a moderator role within a Stack Exchange community has a significant, causal, positive effect on the probability that an individual experiences a purposeful job change. In particular, the probability of a job change increases by between 4.7 and 12.3 percentage points.

This study makes three contributions to the literature. First, our work contributes to the debate on the connection between career benefits and engagement in online communities (Roberts et al. 2006, Hann et al. 2013, Xu et al. 2020). Our study provides, what is to our knowledge, the first causal evidence that community engagement yields concrete career benefits. Our study

is thus in line with prior research, which has suggested that community contributions may have value as a quality signal in the job market (Hann et al. 2013, Huang and Zhang 2016). Second, we extend past work by exploring specific mechanisms behind the observed effects, finding suggestive evidence for a social capital mechanism in particular. Third, we contribute to the literature on individual motivations for contributing to online knowledge communities (see, e.g., Lakhani and Von Hippel 2003, Wasko and Faraj 2005, and Smirnova et al. 2022). Our study suggests that the perceived economic benefits of community engagement, both in general and specifically in terms of moderatorship, are well-founded. Extrinsic, economic incentives are not merely perceived; they are a reality. This has implications for platform operators because economic and social incentives have been shown to have differential impacts on the frequency and quality of user-generated content (Burtch et al. 2018, Kummer et al. 2020). Our work presents an empirical strategy for studying these relationships in other communities and for exploring contingencies of where, how, and for whom career benefits manifest.

2. Theoretical Foundation

2.1. Motivations to Contribute to Online Communities

Knowledge production is increasingly performed by volunteers in online communities, such as those behind question-and-answer forums (Goes et al. 2016, Xu et al. 2020), online encyclopedias (Aaltonen and Seiler 2016, Gallus 2017, Loh and Kretschmer 2023), and open-source software (von Hippel and von Krogh 2003, Maruping et al. 2019). We define online communities (hereafter, communities) as collectives in which volunteers with shared interests participate and interact with the goal of creating an overall outcome mediated by information and communication technology (Armstrong and Hagel 2000).

Extant research focuses on understanding the drivers of contributions to online communities, which is important because an understanding of contributor motivations helps inform the design and management of communities, for example, in terms of incentive mechanisms (Goes et al. 2016, Kokkodis et al. 2020). Previous research documented various drivers, from mere opportunity (Kummer et al. 2020), to different motivations, such as hobbyism (Jeppesen and Frederiksen 2006), the satisfaction of needs (Lakhani and Von Hippel 2003), monetary compensation (Wang et al. 2022), hierarchy (Pu et al. 2022), and peer recognition (Wasko and Faraj 2005, Gallus 2017, Burtch et al. 2022, Smirnova et al. 2022).

These results mirror the labor economics literature, which has considered the motivations for volunteering

(see, e.g., Freeman 1997). Various motives have been observed, including warm-glow altruism (Andreoni 1990, Kotchen and Wagner 2023), charitable giving (Duncan 1999), and tax incentives (Duquette 2016). However, that literature also notes that volunteers may view their contributions as an investment in their future career if the work allows one to acquire new skills, make valuable new contacts, or gain experience (see, e.g., Menchik and Weisbrod 1987).

There are also several relevant studies dealing with the economics of public goods and individual contributions toward public goods. A broad question in that literature pertains to individuals' choice between free-riding and voluntary contributions (see, e.g., Andreoni 1990 and Bénabou and Tirole 2006). In the specific context of online voluntary work, the focus has been on understanding antecedents to contributions to inform the design of platform incentives and interventions (see, e.g., Chen et al. 2010 and Zhang and Zhu 2011). For example, Chen et al. (2010) and Burtch et al. (2018) designed and evaluated experimental interventions that play on social norms.

2.2. Community Leadership

We follow existing research and define a community leader as a member possessing formal authority over a community, in terms of acting in a leader role (Faraj et al. 2015, Johnson et al. 2015, Safadi et al. 2021). The tasks of a leader are to define community-level parameters (e.g., project goal, scope) and to ensure a high quality of community participation. There are different practices by which communities select their leaders, including formal elections, appointment by a council, or more informal approaches, such as *passing the baton* (Burke and Kraut 2008).

Several studies have investigated community leadership (Faraj et al. 2015, Johnson et al. 2015). Their focus is to characterize the traits of users who are perceived as leaders by the community, yet much less attention has been paid to those individuals who hold a formal leadership role. Johnson et al. (2015) and Faraj et al. (2015), for example, explored the behavioral and structural antecedents of leadership in online communities, suggesting that leaders show greater contributions, power, and sociability. Other work has sought to identify predictors of formal leadership; Burke and Kraut (2008) identified factors that influence the likelihood of becoming a Wikipedia admin, and Shah (2006) provided insights on the mechanisms that communities employ to determine their leaders. Yet we still lack a clear understanding of the career benefits one may derive from taking on such a position.

2.3. Career Benefits from Community Engagement

Of relevance to the present study is the role that economic motives may play in driving a desire to lead.

Lerner and Tirole (2002) argued that participating in open-source communities may operate as a springboard for advancing in the corporate world, and they provided some anecdotal evidence. Subsequent studies have documented career benefits as a motivation for community engagement. For example, Roberts et al. (2006) found that paid open-source contributors exhibit higher levels of participation. Xu et al. (2020) observed that Stack Overflow users reduce their intensity of contribution immediately following a job change. And Bitzer and Geishecker (2010) found that university dropouts show a higher probability of contributing to open-source projects, perhaps to improve their career prospects. Kummer et al. (2020) and Malgonde et al. (2023) documented a positive relationship between unemployment and online community contributions. An open question, however, is the extent to which community contributions truly translate into career benefits. At least two studies have provided evidence in this regard. Huang and Zhang (2016) demonstrated that active participation in SAP's online community correlates positively with the likelihood of job-hopping. Hann et al. (2013) found that holding a higher rank in an open-source community correlates positively with wages.

In the wider literature on volunteer work in labor economics, whether and to what extent volunteer work provides career benefits likewise remains an open question. Volunteer work could be perceived negatively by employers, were they to conclude that the volunteering diverts an individual's attention from their career or profession. However, recent evidence suggests some positive career benefit from volunteering (Scharf and Smith 2016, Baert and Vujić 2018, Piopiunik et al. 2020). For example, Piopiunik et al. (2020) found that graduates were more likely to receive a job interview invitation if they had volunteering experience.

2.4. Theoretical Mechanisms

Three theories provide alternative—albeit not mutually exclusive—explanations for how career benefits might manifest: human capital, social capital, and signaling theory.

Human capital theory argues that a worker's marketable skills are capital and that the worker can increase this capital by making investments in skills (Becker 1962). Human capital includes the skills, competencies, and knowledge that individuals can use to increase their productivity. It can explain substantial variation in wage growth (Mithas and Krishnan 2008, Lazear 2009). This capital can take the form of knowledge or any other factor that directly contributes to a worker's productivity. Employment benefits from community participation may therefore derive from gains in human capital. Community participation may create human capital by developing an individual's technical skills,

especially to keep up with technological advances (Zaza et al. 2023). For instance, a user may learn as a result of reviewing computer code or gain an improved understanding from explaining a complex technical concept to others or from engaging in leadership (Lerner and Tirole 2002).

Social capital theory provides an alternative theoretical rationale (Adler and Kwon 2002, Coleman 1994). Social capital refers to “the sum of actual and potential resources embedded within, available through, and derived from, the network of relationships possessed by that individual” (Nahapiet and Ghoshal 1998, p. 243). Together with the human capital view, the social capital view advocates the notion that an individual’s resources are the foundation of career benefits. Individuals with greater social capital may draw on their connections to advance their career. Community engagement is likely to yield social capital because interactions within the community are broadly social and often interpersonal because leaders seldom remain anonymous (Butler et al. 2007, Johnson et al. 2015). These interactions can thus lead to new, meaningful interpersonal relationships, which in turn can advance a career.

Signaling theory is based on the argument that firms have imperfect information about worker productivity, and individuals invest in credentials to credibly convey information about their abilities (Arrow 1973, Spence 1978). Signaling theory argues that employers cannot directly observe worker quality and that a worker’s true quality is only known to him- or herself, that is, the presence of information asymmetries. The informational value of a credential (e.g., a diploma or a degree) comes from employers’ belief that the credential is costly to acquire and positively correlated with a candidate’s true abilities. Community leadership may represent such a signal because firms may infer that an individual holds genuine interest, experience, and endurance.

3. Context and Data

3.1. Study Context

We examine the causal effect of community leadership on career benefits in the context of Stack Exchange, a network of more than 160 online question-and-answer communities. At the time of the study, Stack Exchange collectively hosted more than 20 million questions, with associated answers, contributed by more than 13 million members. Most Stack Exchange communities address IT-related topics (e.g., databases, Unix), but there are also communities focused on non-IT domains (e.g., natural sciences, languages). Each community usually has a group of between two and 30 leaders, so-called moderators (mods).

Mods are expected to contribute expert knowledge and to lead by example (Stack Exchange 2009). Moderators

add or remove community members, review or reject postings, and resolve conflicts. Moderators tend to have an overall greater level of exposure to community members because of their role, and social interactions are a substantial part of their day-to-day activities. Moderatorship may also provide signaling benefits, because an employer may infer that moderators have both knowledge and expertise in their community’s subject matter area. Furthermore, moderatorship can be perceived as useful leadership experience because it bears formal, codified leadership and administrative responsibilities.

Stack Exchange mods are selected through democratic, closed-ballot elections held separately in each community, approximately once a year. Typically, multiple moderator seats are filled in tandem via vote. An election encompasses nomination, primary, and election phases. In the nomination phase, usually seven days long, members can suggest candidates as well as self-nominate. Candidates write a motivation statement and defend themselves in a public Q & A. For example, candidates may be asked about their schedule availability and prior experience. Candidates may also be challenged by members of the community. During the four-day primary, voters can allocate one up- or down-vote per candidate. The top 10 candidates by vote score move to the final election. During the four-day final election, voting follows a single transferrable vote (STV) system, described in Online Appendix B. The elections rely on a secret ballot, the display of the candidates on the ballots is randomized, and safeguards are in place to prevent manipulation. Moderatorship is granted for an unlimited period. Moderator status is indicated by a diamond symbol displayed next to the username.

3.2. Data

We obtain a list of all users that have run for a position as a moderator (*candidates*) in a Stack Exchange community between December 2010, when the first elections were held, and March 2020. We collected data on candidates, seats, votes, and the outcomes from each community’s election web page.³ The initial sample included 1,776 unique candidates across 113 communities and 281 elections. Candidates can run for moderatorship repeatedly, and we keep these repeated candidacies in our sample. To obtain job histories, we manually searched for each candidate’s profile on the professional networking platform LinkedIn. Many candidates disclose their profiles during the nomination phase, along with other personal details. Accordingly, we were able to identify LinkedIn profiles for 65% of all candidates. We retained those candidates who could broadly be considered IT professionals, namely those who held job roles such as software developers, software architects, and data scientists, or

who held degrees with either a major or a minor in computer science. This reduced the sample to 937 candidates. Further details are available in the supplementary files.

3.3. Measures

3.3.1. Career Benefits. We infer career benefits based primarily on the occurrence of job changes. We argue that job changes are a meaningful measure of career benefits for at least three reasons. First, prior research, surveys, and industry reports indicate that job changes among tech workers tend to be career improvements. For example, Topel and Ward (1992) reported that job changes increase worker wages by an estimated 10% on average. Recent surveys and industry reports tell a similar story. Survey results from Pew indicate that most workers who experienced job changes in the last few years experienced wage gains (Pew Research Center 2022). Industry reports indicate that job hopping in the information sector tends to be driven by increased compensation (LinkedIn 2018) and that these job changes result in an approximately 9.7% increase in a worker's wages, a wage benefit larger than that observed in many other industries (Zippia 2023). Second, most job changes are voluntary. Historically, the ratio of *quits* to involuntary separations (layoffs and dismissals) in the United States, otherwise known as the Q/LD ratio, has been greater than 1. This implies that, in general, workers are more likely to leave a job voluntarily than to be fired or laid off (BLS 2018). This is particularly true in the information sector, which frequently exhibited a higher Q/LD ratio than other industries over the period spanning 2000 through 2017. Thus, job changes in the tech sector are very much driven by worker preferences, not firings (BLS 2018). Third, our focus on job changes has the benefit of making our study more directly comparable to prior research (Huang and Zhang 2016, Xu et al. 2020). For example, Huang and Zhang (2016) studied contributions to an online community and similarly relied on job changes as a measure of career improvement. The above points aside, we also validate our assumption that positive effects on job changes amount to career benefits by considering salary data as an alternative measure, as we describe below. To identify job changes, we extract candidates' job history from their LinkedIn profile and code breakpoints (*Job change*). Here, we include job changes within the same company, and we also include job changes with intermediary employment gaps.

We also consider an alternative measure, namely an estimate of a candidate's job salary (*Salary*). To construct this measure, we first standardize the job roles reported in candidates' job histories (e.g., *Senior SDK Developer* to *Senior Developer*) and then obtain the average annual salary for that job role within the employing company from GlassDoor, an online platform where employees

share salary information. Whenever company-role-specific data were not available, we imputed it using data on the same job role at the smallest available geographical area (i.e., city or region, country). Because not all candidates reported their employing company or region, salary data are available only for a subset of individuals.

3.3.2. Treatment. *Mod* is 1 if candidate *i* was elected as a moderator.

3.3.3. Education and Career-Related. Based on the LinkedIn profiles, we create indicator variables that capture different levels of educational attainment of a candidate, that is, undergraduate degree and doctoral degree. We also create career-related variables. *Workxp (yrs)* is *i*'s total number of years in employment, and we round the variable to 1 when a candidate has less than one year of work experience. *Workxp (jbs)* is *i*'s total number of jobs. Both variables are measured for the pre-election period.

3.3.4. Participation. We also create variables capturing candidate *i*'s participation in the community. *Answers* is the number of answers contributed by *i*; *Questions* captures the number of questions submitted by *i*; *Edits* indicates the number of posts by other users that were edited by *i*; *Badges* captures the count of badges (i.e., awards for participation) that *i* has received; *Comments* is the number of comments posted by *i*; and *Suggestions* captures the count of suggestions for edits of others' posts that *i* has made. These variables—unless noted otherwise—are observed over the half-year before a candidate's election date for the purpose of assessing pre-election heterogeneity. In the few cases where a candidate runs multiple times in an election, the variable holds the data for the respective half-year before the current election. Each of these variables is logged to address right skewness.

3.3.5. Mechanisms. We proxy for human capital acquisition following an election by considering a candidate's subsequent accumulation of knowledge tags in the community. Stack Exchange relies on a system of tags to categorize questions (e.g., *Python*, *Android*). The more tags a candidate is involved in, the more human capital accumulation we might infer. Tag badges are automatically conferred to users if their activity exceeds a certain threshold of activity (e.g., providing answers or edits) around posts on a specific tag. *Tag badges* is the count of tag badges acquired by candidate *i* over the second half-year following the election (i.e., $t + 1$).⁴ Note that acquisition of these badges plausibly reflects only human capital accumulation; these badges are not likely to provide any signaling benefits because, unlike the receipt of a diploma or degree, the effort and skill involved in

acquiring these badges is nonstandard and hard for an employer to infer, given that the badge names are unintuitive for employers (e.g., “Necromancer”, “Lifejacket”). We infer social capital from the number of individuals a candidate has interacted with in the Stack Exchange chat. Chats hint at interactions between users and provide an understanding of the social capital of a user. Stack Exchange relies on a public chat room system. Whereas users can create private rooms (i.e., rooms that are accessible by invitation only), all other messages are publicly available along with timestamps. We create the variable *Chat contacts* as the number of users who had sent a direct message to candidate i ; consistent to the human capital measure, we capture the activity over the second half-year following the election (i.e., $t + 1$). We count only incoming messages, because outbound messages may reflect mere interest in forming connections, rather than actual reciprocal communication.⁵ We log transform both measures.

3.3.6. Community and Election. Stack Exchange reports community and election data on the election website of each community. *Community size* is equal to the number of unique, eligible voters who visited the forum between the beginning of the nominations and the end of the election. A unique visit refers to the fact that a voter opened the forum website, regardless of how often they visited. We log-transform the measure. We define *Election participation* as the number of unique, eligible voters who visited the forum during the election phase (i.e., *Community size*), divided by the total number of eligible voters as reported on the election webpage of each community. Higher participation in the election may lead to increased awareness and transparency, which in turn may influence election outcomes. Data on the number of eligible voters is reported on the Stack Exchange election pages. *Moderators* is the ratio of the number of moderator positions to *Community size*. This ratio may hint at the moderators’ workload. We create dummies for the community subject-matter theme, as derived from the official categorization provided by Stack Exchange at the time of the study, namely business (e.g., Project Management), culture (e.g., German language), life (e.g., Gardening & Landscaping), professional (e.g., Freelancing), science (e.g., Biology), and technology (e.g., Stack Overflow). We also add several variables that the communities display to voters about i as information about the merits of a candidate. *Reputation* holds the reputation score of i at the time of the nomination. In addition, the candidate display includes information about the achievements of certain badges. Therefore, complementary to the general *Badges* variable, we capture these specific badge types in separate variables. *Editing badges* is the count of badges received by i for editing activity (i.e., given to honor contributions with regard to editing posts). *Moderation badges* is the count of badges for

moderation activity received by i (i.e., given to honor contributions that help the actual moderators, such as flagging posts). *Participation badges* is the count of badges for participation activity received by i (e.g., badges given for completing the user profile description or regular visits of the community). To adjust for community-specific differences in the reputation distribution as well as the requirements to obtain badges, we normalize these variables by *Community size*, and we log-transform them to account for right skewness.

Table 1 presents the variables used in the study and their descriptive statistics. Not reported in the table but worth noting to understand the sample is that the most frequent job roles are software developer or engineer (12.58%), consultant (8.96%), web developer (7.12%), engineer (6.94%), and technician (5.43%).

4. Estimation and Identification Strategy

4.1. Hazard Framework

Following research on labor mobility and to account for the fact that job change events are distributed in time and are right censored for all candidates who do not change employers by the end of the observation window, we employ a hazard model (see, e.g., Card et al. 2007, Chetty 2008, Yang 2017).⁶ Hazard models are preferable when the focus is on understanding the relationship between the *risk* of an event occurring at time t and the explanatory variables of interest at the same point in time (Lancaster 1990). We employ a Cox proportional hazard (PH) model because, unlike other hazard approaches, it does not impose any assumptions on the functional form of the baseline hazard rate (Cox 1972, Lancaster 1990).

4.2. Identification Strategy

We employ two distinct yet complementary identification strategies: a difference-in-differences design (DID) with matching and a regression discontinuity design (RDD). The DID design compares the occurrence of job changes before versus after elections, contrasting elected and unelected candidates. The DID has the dual benefit of yielding an ATE and helping to account for unobserved time-invariant heterogeneity between candidates via the incorporation of candidate fixed effects in addition to matching. The (sharp) RDD is a quasi-experimental design that allows the identification of a LATE when treatment assignment changes discontinuously at a threshold. Following the ample literature on the use of RDD in studying electoral races, we compare the job outcomes of candidates within a neighborhood (*bandwidth*) around the winning threshold (Imbens and Lemieux 2008, Lee and Lemieux 2010, Eggers et al. 2015). The running (forcing) variable in this setup is the candidate’s vote share margin, and the cutoff is the vote threshold at which a candidate is

Table 1. Variables

Variable	Description	Mean	SD	Min.	Med.	Max.
<i>Job change</i> ^A	1 if candidate <i>i</i> reported a job change	0.51	0.50	0.00	1.00	1.00
<i>Job change (frequency)</i> ^A	Total no. of job changes by candidate <i>i</i> divided by <i>Workxp (yrs)</i>	0.58	0.43	0.00	0.50	4.00
<i>Log(Salary)</i> ^A	(Annual) estimated salary of candidate <i>i</i> in USD	11.44	0.40	9.92	11.43	12.44
<i>Mod</i> ^T	1 if candidate <i>i</i> was elected as moderator	0.34	0.48	0.00	0.00	1.00
<i>Undergraduate degree</i> ^B	1 if candidate <i>i</i> reported an undergraduate degree	0.34	0.47	0.00	0.00	1.00
<i>PhD</i> ^B	1 if candidate <i>i</i> reported a PhD degree	0.07	0.25	0.00	0.00	1.00
<i>Workxp (yrs)</i> ^B	Work experience in years of candidate <i>i</i>	10.24	7.50	1.00	8.83	40.58
<i>Workxp (jbs)</i> ^B	Work experience as job count of candidate <i>i</i>	4.88	4.03	0.00	4.00	38.00
<i>Log(Answers)</i> ^{T-1}	Answers posted by candidate <i>i</i>	11.02	8.21	0.00	10.64	34.19
<i>Log(Questions)</i> ^{T-1}	Questions posted by candidate <i>i</i>	2.21	3.14	0.00	0.69	20.14
<i>Log(Edits)</i> ^{T-1}	Edits of posts by candidate <i>i</i>	13.37	9.76	0.00	13.17	44.71
<i>Log(Badges)</i> ^{T-1}	Total badges received by candidate <i>i</i>	7.80	4.79	0.00	7.68	24.49
<i>Log(Comments)</i> ^{T-1}	Comments posted by candidate <i>i</i>	15.37	9.99	0.00	15.95	42.16
<i>Log(Suggestions)</i> ^{T-1}	Suggestions reviewed by candidate <i>i</i>	1.31	2.96	0.00	0.00	23.47
<i>Log(Tag badges)</i> ^{T+1}	Tag badges awarded to candidate <i>i</i>	0.66	1.17	0.00	0.00	6.58
<i>Log(Chat contacts)</i> ^{T+1}	Count of users who had sent a direct message to candidate <i>i</i>	0.12	0.31	0.00	0.00	1.46
<i>Log(Community size)</i> ^{N, C}	No. of unique eligible voters who visited the forum during the election phase	6.93	1.95	3.74	6.46	12.37
<i>Election participation</i> ^{N, C}	<i>Community size</i> divided by the total no. of eligible voters	0.38	0.12	0.11	0.36	0.90
<i>Moderators</i> ^{N, C}	Number of moderator positions in the community, divided by <i>Community size</i>	0.01	0.01	0.00	0.00	0.07
<i>Log(Reputation)</i> ^N	Reputation score of candidate <i>i</i> as displayed during the nomination phase, divided by <i>Community size</i>	1.30	1.97	-8.14	1.76	4.66
<i>Log(Editing badges)</i> ^N	Editing badges awarded to candidate <i>i</i> as displayed during the nomination phase, divided by <i>Community size</i>	0.00	0.00	0.00	0.00	0.04
<i>Log(Mod. badges)</i> ^N	Moderation badges awarded to candidate <i>i</i> as displayed during the nomination phase, divided by <i>Community size</i>	0.00	0.00	0.00	0.00	0.03
<i>Log(Partic. badges)</i> ^N	Participation badges awarded to candidate <i>i</i> as displayed during the nomination phase, divided by <i>Community size</i>	0.00	0.01	0.00	0.00	0.08
<i>Community theme (%)</i> ^C						
Business	0.53	Professional				1.81
Culture	15.15	Science				6.51
Life	9.61	Technology				66.38

Notes. N = 937 candidates; A, for the postelection period; B, for before the election; C, measured at the level of the community/election; N, measured at the time of the nomination; T, measured at the time of the election; T + 1, measured in *t* + 1; T - 1, measured in *t* - 1.

Table 2. Pre-Election Balance Before and After Matching (DID)

	Panel A: DID-All (N = 937)			Panel B: DID-CEM (N = 319)			Panel C: DID-PSM (N = 587)		
	W. Mean	N.-W. Mean	Diff.	W. Mean	N.-W. Mean	Diff.	W. Mean	N.-W. Mean	Diff.
<i>Job change</i>	0.56	0.60	−0.03	0.62	0.51	0.11	0.59	0.62	−0.03
<i>Job change (frequency)</i>	0.53	0.60	−0.07**	0.54	0.51	0.03	0.55	0.58	−0.02
<i>Undergraduate degree</i>	0.39	0.32	0.07**	0.40	0.34	0.06	0.40	0.37	0.04
<i>PhD</i>	0.10	0.05	0.05***	0.08	0.08	0.00	0.09	0.08	0.02
<i>Workxp (yrs)</i>	11.81	9.42	2.39***	12.02	11.26	0.76	11.32	10.64	0.69
<i>Workxp (jbs)</i>	5.11	4.75	0.36	5.38	5.39	−0.01	5.06	4.85	0.20
<i>Log(Answers)^M</i>	12.96	10.06	2.90***	9.30	9.55	−0.24	11.97	13.03	−1.06
<i>Log(Questions)^M</i>	2.24	2.20	0.04	1.03	0.96	0.07	2.31	2.04	0.26
<i>Log(Edits)^M</i>	16.33	11.89	4.44***	11.81	11.93	−0.12	14.97	16.07	−1.09
<i>Log(Badges)^M</i>	8.85	7.29	1.56***	6.74	6.92	−0.18	8.33	8.54	−0.21
<i>Log(Comments)^M</i>	17.95	14.09	3.86***	13.76	13.61	0.15	16.70	17.96	−1.27
<i>Log(Suggestions)^M</i>	0.91	1.52	−0.61***	0.49	0.64	−0.15	1.11	0.89	0.22
<i>Log(Tag badges)^M</i>	0.41	0.21	0.20***	0.22	0.28	−0.05	0.32	0.39	−0.06
<i>Log(Chat contacts)^M</i>	0.25	0.12	0.13**	0.14	0.12	0.02	0.18	0.21	−0.03
<i>Log(Reputation)^M</i>	2.13	0.86	1.27***	1.82	1.86	−0.05	1.83	1.75	0.08
<i>Log(Editing badges)^M</i>	0.01	0.00	0.00***	0.00	0.00	0.00	0.00	0.00	0.00
<i>Log(Mod. badges)^M</i>	0.01	0.00	0.00***	0.00	0.00	0.00	0.00	0.00	0.00
<i>Log(Partic. badges)^M</i>	0.01	0.00	0.00***	0.00	0.00	0.00	0.01	0.01	0.00

Notes. Panel B and C report means weighted by the CEM and PSM weights, respectively. Difference is a regression estimate. N reports the number of candidates. Means and differences are rounded to the second digit. W., winner; N.-W., nonwinner; M, matching variable.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

elected. Compared with a DID design, an RDD rests on fewer assumptions when it comes to identification, because it relies to a lesser degree on the direct observation of all relevant covariates (Angrist and Pischke 2009). The downside of the RDD is that it yields only a LATE, and furthermore, it focuses on a smaller, pruned sample (Imbens and Lemieux 2008).

4.3. Difference-in-Differences Design

4.3.1. Panel Construction and Matching Techniques. Like Chetty (2008) and Marinescu (2009), we construct our data set for the DID estimation as an employment spell panel. We consider a pre-election and post-election period of two years around each election, to ensure that we observe a lengthy period of pre-election activity, while also ensuring a reasonably large estimation sample. This means that we split the employment history of a candidate into spells that begin with a job change and end with the subsequent job change. We then transform the spells with the remainder of the data into a half-year panel such that a spell can begin or end with only a half-year. We follow the approach of Cengiz et al. (2019), aligning elections in relative time.⁷ To reduce heterogeneity, we employ propensity score matching (PSM) and coarsened exact matching (CEM). Regarding PSM, we employ logistic nearest-neighbor matching (Rosenbaum and Rubin 1983). Regarding CEM, we define the cut points manually to obtain balance in post-matching tests of mean differences (Iacus et al. 2011). For both procedures, we match on the following

variables; for the time-variant variables based on the half-year preceding an election: *Log(Answers)*, *Log(Questions)*, *Log(Edits)*, *Log(Badges)*, *Log(Comments)*, *Log(Suggestions)*, *Log(Reputation)*, *Log(Editing badges)*, *Log(Moderation badges)*, *Log(Participation badges)*, *Log(Tag badges)*, and *Log(Chat contacts)*.⁸ Matching on these variables ensures the comparability of candidates in terms of their community contributions and the performance criteria that are publicly displayed about candidates during the election. We do not match on pretreatment *Job change*, because recent work notes that this can introduce bias into the estimation (Ding and Li 2019). This procedure yields three panels, one based on CEM (hereafter DID-CEM), one based on PSM (hereafter DID-PSM), and one that includes all candidates, that is, without matching (hereafter DID-All).

Table 2 compares means between winners and non-winners in (Panel A) DID-All, (Panel B) DID-CEM, and (Panel C) DID-PSM. Considering Panel A, we observe significant differences on several variables. These differences are eliminated by the matching techniques, as shown in Panels B and C.

4.3.2. Regression Specification. Following Chetty (2008) and Marinescu (2009), the DID then takes the form

$$h(t) = h_0^i(t) \exp(\beta_1 \text{After}_t \times \text{Mod}_i + \phi_i) \quad (1)$$

where candidate i is observed in half-year t , *After* is 1 for half-years after the election, *Mod* defined as above,

and ϕ includes dummies specific to each half-year relative to the election. The β_1 captures how intertemporal changes in the hazard of job change (before versus after an election) differ between winners and nonwinners. To avoid incidental parameter bias, and violation of the base assumption of the proportional hazard specification, candidate-fixed effects are implemented via stratification (Allison 2009, chapter 5). To adjust for the panel structure, we cluster the standard errors at the employment spell level (Abadie et al. 2023).

4.4. Regression Discontinuity Design

4.4.1. Majority Threshold and Sample Construction. The sample construction for the RDD depends on the election system. Stack Exchange elections rely on a preference-based, single-transferrable vote system (STV). In an STV system, voters name and rank their top three candidates on a ballot. Votes are then iteratively reallocated based on voters' candidate rankings. In each round, a quota is fixed. If a candidate's votes reach that quota, then that candidate is elected. Any votes exceeding the quota are redistributed based on the second and third preferences on the ballot. If none of the remaining candidates surpass the current quota in a given iteration, then the lowest-performing candidate is eliminated, and the excess votes for the candidate are again redistributed. Most elections involve several rounds of vote reallocation. Given this setup, in STV systems, elections are not won on the basis of a simple majority of initial votes (i.e., 50%); instead, each iteration produces a distinct election threshold (*quota*).

To adjust the sharp RDD setup to the STV system, we follow the approach of Redmond and Regan (2015); to obtain a meaningful majority cutoff, the sample is restricted to the pair of candidates who contested the final available seat in an election and after all vote transfers were made. For each final seat, one candidate will ultimately win (*least-voted winner*), and one will be the runner-up (*most-voted loser*). If candidates i and j contest the final seat, then the candidate who receives more than 50% of the remaining votes will become a moderator. Formally, if candidates i and j contest the final seat, then the equivalent vote share for i is thus calculated as the votes for i divided by the sum of votes for i and j . We exploit the sharp discontinuity at the 50% threshold to estimate the change in the dependent variable with assignment to *Mod*. Let Y_i denote the dependent variable, and then we are interested in the discontinuity in the conditional expectation of Y_i given the vote share, to identify the causal effect, c , at the discontinuity. Candidate i is elected (*Mod*) if i 's vote share is above 50% in the final round of vote reallocation. With *Share margin*, we accordingly denote the vote share of candidate i minus the winning threshold of 50%.

This RDD is based on a cross-sectional sample of candidates (instead of a candidate panel). That is, our estimation sample is comprised of one observation per candidate, and the outcome of interest is the duration of time to their next observed job change, with right censoring at the end of our observation window for candidates where no job change occurs. For a few elections, we observe only one final seat candidate (i.e., either the least-voted winner or the most-voted loser). We retain those elections in the sample because the RDD does only require probabilistic equivalence of the candidates. Moreover, as we document in the robustness section, there is no evidence of a systematic bias in data availability depending on the election outcome.

4.4.2. Regression Specification. We follow extant studies that used RDDs under hazard modeling frameworks (see, e.g., Berk and de Leeuw 1999 and Card et al. 2007) and define the model as

$$h(t) = h_0(t) \exp(\alpha + \beta_0 \text{Mod}_i + \beta_1 \text{Share margin}_i + \beta_2 \text{Mod}_i \times \text{Share margin}_i + \beta_3 x_{i1} + \dots + \beta_k x_{ik}) \quad (2)$$

where $h(t)$ is the hazard function, and h_0 is the baseline hazard of job change (i.e., the instantaneous hazard of job change among nonwinners). Survival time is measured in half-years. The specification follows the local pooled regression (Lee and Lemieux 2010). The interaction term allows the relationship between the running variable and the outcome to differ on either side of the threshold. We decided for this specification based on Bor et al. (2014) but also the more general advice to estimate a variety of models (Lee and Lemieux 2010) and considering the Akaike information criterion (AIC). This approach is asymptotically equivalent to a selection based on leave-one-out, cross-validated (LOOCV) R^2 (Stone 1977). The x 's in this specification represent a set of k controls ($x_1, x_2, \dots, x_k$) along with their respective coefficients (in this case $\beta_3, \beta_4, \dots, \beta_k$). The controls are *Undergraduate degree* and *PhD*, to control for the educational background of the candidate (Huang and Zhang 2016); *Community size*, to adjust for the potential competition for the moderator position (Loh and Kretschmer 2023); *Election participation*, because election participation may lead to increased awareness and transparency of candidates, which in turn may influence election outcomes; *Moderators*, to adjust for the moderation workload necessary in a community; *Reputation*, to control for the quality or expertise of a candidate (Goes et al. 2016, Smirnova et al. 2022); and *Community theme*, to adjust for factors correlated with the different types of communities.⁹

4.4.3. Bandwidth Selection and Comparability. The RDD is ultimately estimated on a restricted subset of candidates that lie within the neighborhood of the vote threshold, defined in terms of a bandwidth parameter. To select the bandwidth, we rely on the bandwidth selection algorithm of Imbens and Kalyanaraman (2012). The algorithm is based on minimizing an expected squared error loss criterion, which incorporates several factors, including the variance in outcomes at the cutoff, the density of the running variable, covariates, and the shapes of the curves on both sides. The idea of the criterion is that a larger bandwidth should be chosen if the variance in outcomes increases at the cutoff, when the density of the running variable is small, and the shapes of the curves at either side of the cutoff become symmetrical. The obtained bandwidth—in the following referred to as *IK-bandwidth*—is 15.9%. Recognizing that the IK-bandwidth in our case may be perceived as large, we report the results also for a bandwidth of 5%. It yields a large-enough sample to permit statistical tests but is considerably smaller than the IK-bandwidth. This procedure results in three samples, one containing all final seat candidates (hereafter, RDD-None), one containing final seat candidates within the IK-bandwidth (hereafter RDD-IK), and one containing final seat candidates within the 5% bandwidth (hereafter RDD-5%).

Table 3 tests for pre-election differences in means between winners and nonwinners within our RDD samples, namely (Panel A) RDD-None, (Panel B) RDD-IK,

and (Panel C) RDD-5%. None of these samples shows a statistically significant difference in job changes. Whereas we observe imbalance along some variables in RDD-None, we observe no significant imbalance in RDD-IK.

Figure 1 shows complementary discontinuity plots for RDD-None for visual inspection of the job change and participation variables, which confirm what the statistical analysis has indicated. Online Appendix Table A.1 compares DID-All versus RDD-None and RDD-None versus RDD-IK.

5. Results

5.1. Model-Free Evidence

Figure 2 plots model-free evidence of career benefits in terms of the Kaplan-Meier failure curves for the different subsamples, where the vertical axis is the cumulative fraction of failures (i.e., job change) and the horizontal axis reflects half-years after the election. The solid curve denotes winners, the dashed curve nonwinners. The difference between these two curves reflects the hazard ratio. This figure shows that within the group of winners, a higher fraction of candidates experiences a job change compared with candidates within the group of nonwinners. In all plots, the differences are marked over the entire period of analysis. Taken together, these results show that the group of winners exhibits job changes at higher rates and earlier compared with the group of nonwinners.

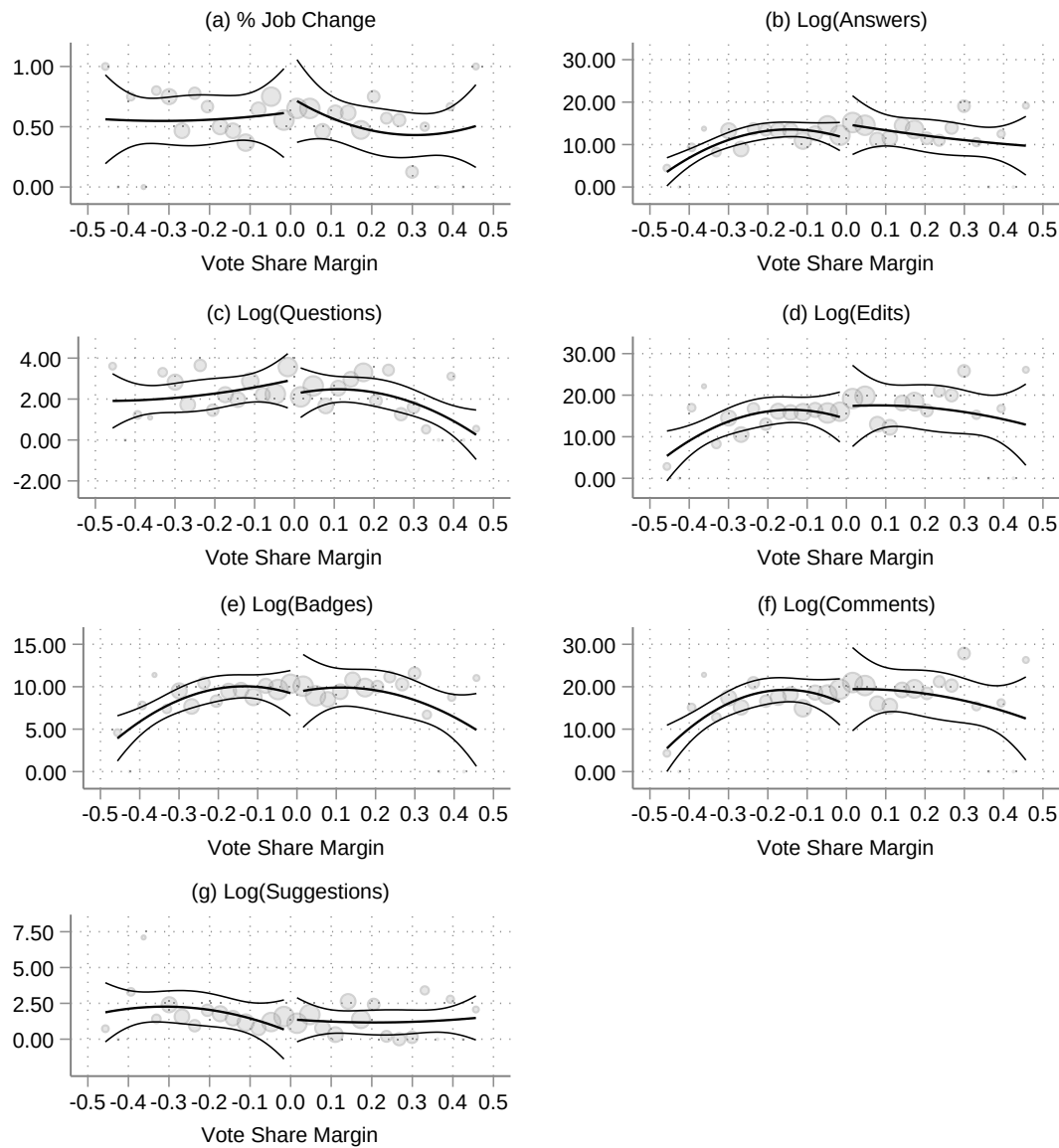
Table 3. Pre-Election Balance Between Winners and Non-Winners (RDD)

	Panel A: RDD-None (N = 323)			Panel B: RDD-IK (N = 187)			Panel C: RDD-5% (N = 77)		
	W. Mean	N.-W. Mean	Diff.	W. Mean	N.-W. Mean	Diff.	W. Mean	N.-W. Mean	Diff.
<i>Job change</i>	0.57	0.60	0.03	0.60	0.57	−0.03	0.61	0.65	0.04
<i>Job change (frequency)</i>	0.53	0.58	0.06	0.54	0.55	0.01	0.52	0.63	0.10
<i>Undergraduate degree</i>	0.38	0.31	−0.07	0.41	0.36	−0.05	0.44	0.40	−0.04
<i>PhD</i>	0.09	0.06	−0.03	0.07	0.07	0.00	0.03	0.05	0.02
<i>Workxp (yrs)</i>	11.81	10.11	−1.70*	11.39	11.19	−0.20	10.48	9.19	−1.29
<i>Workxp (jbs)</i>	5.08	4.85	−0.24	5.33	4.88	−0.45	4.44	4.53	0.08
<i>Log(Answers)^M</i>	13.73	12.40	−1.33	13.68	12.98	−0.69	15.48	13.83	−1.65
<i>Log(Questions)^M</i>	2.35	2.58	0.23	2.42	2.73	0.31	2.34	3.10	0.76
<i>Log(Edits)^M</i>	18.19	15.06	−3.13***	17.22	16.26	−0.96	19.54	16.19	−3.35
<i>Log(Badges)^M</i>	9.77	9.38	−0.40	9.59	9.89	0.31	9.83	10.25	0.43
<i>Log(Comments)^M</i>	19.61	17.45	−2.15**	18.93	18.24	−0.69	21.23	19.63	−1.60
<i>Log(Suggestions)^M</i>	1.30	1.54	0.24	1.36	1.28	−0.08	1.01	1.42	0.42
<i>Log(Tag badges)^M</i>	0.46	0.29	−0.17*	0.47	0.32	−0.15	0.48	0.32	−0.16
<i>Log(Chat contacts)^M</i>	0.32	0.19	−0.13	0.33	0.14	−0.18	0.42	0.11	−0.31
<i>Log(Reputation)^M</i>	2.13	1.57	−0.56***	2.11	1.73	−0.38	1.49	1.38	−0.11
<i>Log(Editing badges)^M</i>	0.00	0.00	0.00**	0.01	0.00	0.00	0.00	0.00	0.00
<i>Log(Mod. badges)^M</i>	0.00	0.00	0.00***	0.00	0.00	0.00	0.00	0.00	0.00
<i>Log(Partic. badges)^M</i>	0.01	0.01	0.00	0.01	0.01	0.00	0.01	0.01	0.00

Notes. N reports the number of candidates. Means and differences are rounded to the second digit. Difference is a regression estimate. W., winner; N.-W., nonwinner; M, matching variable.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

Figure 1. Pre-Election Regression Discontinuity Plot of Job Change and Participation Outcomes



Notes. This figure plots pre-election outcomes by candidate bins of the vote share margin on the horizontal axis. The markers to the right are bins of elected winners (treatment group), and those to the left are bins of nonwinners (control group). Marker size indicates the number of candidates in the bin. Sample is RDD-None. We employ the default bin sizes in the cmogram function in Stata. The curve is a quadratic fit to either side of the threshold along with 90% confidence intervals.

5.2. Difference-in-Differences Evidence

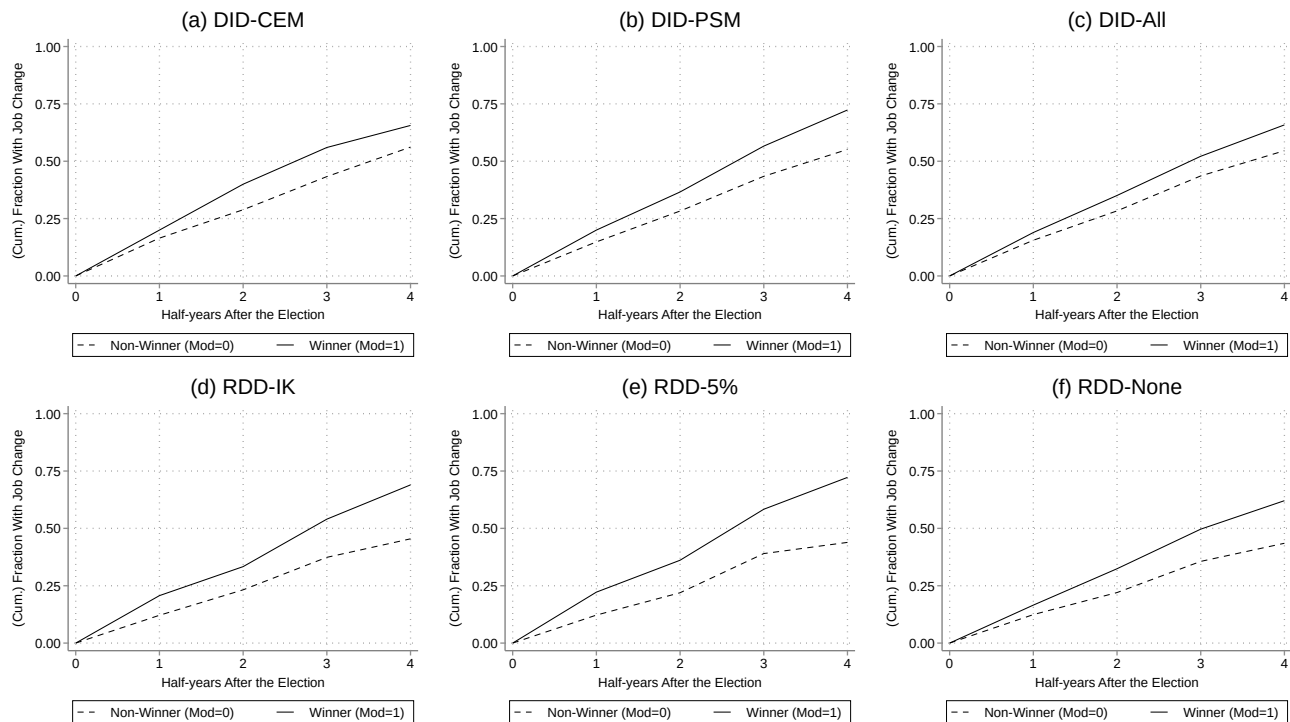
Table 4 reports the results for the DID in terms of Equation (1). The reported coefficients represent hazard ratios (i.e., exponentiated, nonnegative coefficients, centered on 1). Coefficients below 1 represent declines in the hazard of job change, whereas coefficients above 1 represent an increase in the hazard of job change. Column (1) reports estimates for DID-CEM. The coefficient associated with the DID term is positive and significant, indicating that winners exhibit a significantly higher hazard of job change following their election as compared with CEM-matched nonwinners (65.3%). Column (2) reports

the estimates for DID-PSM, yielding a positive and significant effect (89.5%). Finally, column (3) reports estimates DID-All, further confirming a significant and positive effect (23.3%).

5.3. Regression Discontinuity Evidence

Figure 3 shows model-free evidence in terms of the percent of candidates in RDD-None per vote share bin who experience a job change over the postelection period against their vote share margin. The curves are fit separately on either side of the threshold together with 90% confidence intervals. The plot indicates a marked

Figure 2. Model-Free Evidence of Moderatorship and Job Changes



Notes. The graphs show the Kaplan-Meier failure curve for the respective subsamples, where the vertical axis is the failure (i.e., cumulative fraction/percent of candidates with job change) and the horizontal axis gives the half-years after the election. The solid curve is for winners, the dashed curve for nonwinners. Panel (a) is DID-CEM. Panel (b) is DID-PSM. Panel (c) is DID-All. Panel (d) is RDD-IK. Panel (e) is RDD-5%. Panel (f) is RDD-None.

discontinuity at the vote share threshold, which suggests that moderatorship has a positive effect on job prospects.

Table 5 reports the RDD estimates. The coefficients are hazard ratios (i.e., exponentiated, nonnegative coefficients where a null result is implied by an estimate of 1). Column (1) reports the estimates from Equation (2), without controls, for RDD-IK. The estimate of the discontinuity is positive and significant, indicating that being elected as a moderator increases the hazard of job change (97.9%). Column (2) uses the same setup but adds controls, and the coefficient of interest is positive and significant (117.4%). Column (3) uses RDD-5%

(432.8%). Column (4) uses RDD-None (78.9%). Taken together, the RDD results show that being elected as a moderator increases the hazard of job change.

Figure 4 shows the estimated hazard ratios from Equation (2) for various bandwidths to ensure that the results are not driven by arbitrary choices of the bandwidth. We consider bandwidths between 1% and 50%, centered at the threshold, by increments of 0.25 percentage points. The black dots reflect the hazard ratio associated with the coefficient. The vertical line indicated as *IK* refers to the Imbens and Kalyanaraman (2012) bandwidth (i.e., as used for the RDD-IK sample), and 5% refers to the 5% bandwidth (i.e., as used for the

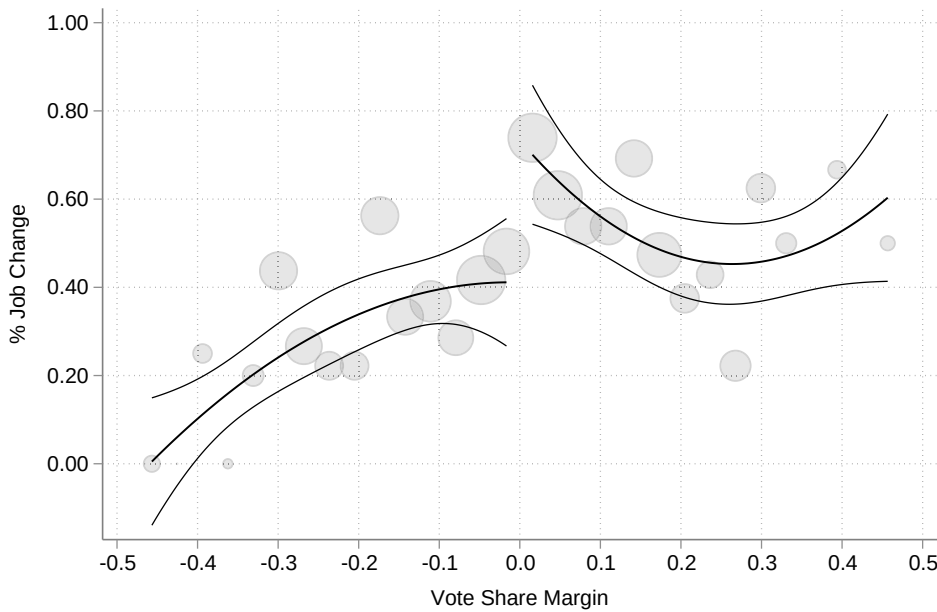
Table 4. Difference-in-Differences Estimates of Job Change Hazards

	DV = Job change		
	(1)	(2)	(3)
After × Mod	1.653** (0.396)	1.895*** (0.366)	1.233* (0.155)
Sample	DID-CEM	DID-PSM	DID-All
N (candidates/observations)	319/2,502	585/4,588	932/7,297

Notes. Cox PH model estimates. Coefficients are reported as hazard ratios. Observations are employment-spells observed every half-year. Candidates not observed after the election are omitted from the estimation. Standard errors are in parentheses. Coefficients and standard errors are rounded to three decimal places.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

Figure 3. Regression Discontinuity Plot



Notes. This figure plots the percent of candidates per vote share bin who experience a job change over the postelection period against their vote share margin. The markers are candidate bins. Marker size indicates the number of candidates in the bin. Sample is RDD-None. We employ the default bin sizes in the `cmogram` function in Stata. The curve is a quadratic fit to either side of the threshold along with 90% confidence intervals.

RDD-5% sample). The error bars reflect 90% confidence intervals. Panel (a) reports results without controls; panel (b) includes controls.

5.4. Summary of Results and Interpretation

Table 6 compares the estimated hazard ratios across the models. In the DID estimation, the hazard ratios range from 1.233 to 1.895. In the RDD estimation, the hazard ratios range from 1.789 to 5.328, and they are larger for smaller bandwidths around the cutoff. The remaining columns report the predicted probability of experiencing a job change over the postelection period for winners and nonwinners. The *Difference*-column reports the difference between the predicted probabilities for winners and nonwinners. Being elected as a moderator increases, on average, the probability of a job change by between 4.7 to 12.3 percentage points.

There are only a few estimates available for comparison. Notably, Huang and Zhang (2016) found that a 1% increase in knowledge contributions (measured by the total helpful answers posted by users in an enterprise software community) is associated with a 1.2 percentage point increase in the probability of a job change. Our estimate is higher, indicating that leadership is associated with significantly greater career benefits than knowledge contributions. However, it is important to note that these estimates are not perfectly comparable.

Two patterns in the estimates are important to be discussed.¹⁰ The first pattern is that the effect magnitude

for the smallest bandwidth (i.e., RDD-5%) is larger than for RDD-IK and RDD-None. A similar pattern is also evident in other RDD studies (see, e.g., Ludwig and Miller 2007, Flammer 2015). One explanation is that the number of observations decreases with smaller bandwidths, thus making sampling variability a more important part of the estimate (see Lee and Lemieux 2010 and Calonico et al. 2014). Indeed, this issue motivates using the Imbens and Kalyanaraman (2012) bandwidth because it considers the variance in the outcome at the cutoff. Another explanation is also possible, namely that the baseline hazards in the respective subsamples differ. The second pattern is that the smallest effect magnitude is the one obtained from DID-All. One interpretation is that the endogeneity—which enters the estimation as we move from DID-All to the matched samples (i.e., DID-CEM and DID-PSM) or the discontinuity samples (i.e., RDD-IK, RDD-5%, RDD-None)—exerts a downward bias. As a result, the effect on the outcome variable may be underestimated in this model, leading to a smaller coefficient estimate. Arguably, among the estimated models, DID-All is one most prone to endogeneity because it is not adjusted to the pre-existing heterogeneity between winners and nonwinners documented in Table 2 and because it does not account for unobserved variables negatively correlated with winning. Another interpretation is that DID-All provides an estimate of the ATE, whereas the matched and RDD estimates provide a LATE. The matching and RDD setups could shift the observed job changes to a

Table 5. Regression Discontinuity Estimates of Job Change Hazards

	DV = <i>job change</i>			
	(1)	(2)	(3)	(4)
<i>Mod</i>	1.979** (0.678)	2.174** (0.727)	5.328*** (2.461)	1.789** (0.452)
<i>Share margin</i>	5.641 (19.087)	0.528 (1.933)	0.000 (0.000)	1.181 (1.448)
<i>Mod × Share margin</i>	0.019 (0.083)	0.191 (0.923)	0.000 (0.000)	0.448 (0.796)
Controls				
<i>Undergraduate degree</i>		1.357 (0.278)	1.653 (0.558)	1.578*** (0.270)
<i>PhD</i>		0.444 (0.290)	0.910 (1.626)	0.327** (0.170)
<i>Log(Community size)</i>		1.307 (0.316)	2.571 (1.492)	1.175 (0.161)
<i>Election participation</i>		2.356 (2.800)	0.199 (0.437)	2.240 (2.014)
Moderators		0.987 (0.016)	0.976 (0.023)	0.993 (0.012)
<i>Log(Reputation)</i>		1.246 (0.289)	2.520 (1.455)	1.107 (0.139)
<i>Community theme - culture</i>		0.062*** (0.025)	—	0.158*** (0.081)
<i>Community theme - life</i>		0.126*** (0.053)	3.518* (2.390)	0.209*** (0.111)
<i>Community theme - professional</i>		0.000*** (0.000)	—	0.062** (0.068)
<i>Community theme - science</i>		0.118*** (0.064)	4.959** (3.749)	0.154*** (0.088)
<i>Community theme - technology</i>		0.087*** (0.034)	2.378 (1.278)	0.166*** (0.084)
Sample	RDD-IK	RDD-IK	RDD-5%	RDD-None
<i>N</i> (candidates)	187	187	77	323

Notes. Cox PH model estimates. Coefficients are reported as hazard ratios. *N* gives the sample size in candidates. Standard errors in parentheses. Coefficients and standard errors are rounded to three decimal places. Coefficients marked with “—” not estimated due to no variation in these variables in the respective subsample.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

subgroup of individuals for whom the career benefits from moderatorship are larger than average. In Table 2, the group means indicate that the matching and RDD setups tend to discard candidates from the sample who participated most actively before the election. For example, highly participating candidates could experience relatively smaller treatment effects, perhaps because they have already accumulated considerably human and social capital or because they possess signals other than moderatorship.

5.5. Further Evidence: Salary Estimation

To bolster the interpretation of job changes as career benefits, we next consider wage effects. Tables 7 and 8 report the results, which support a career benefit interpretation. For the DID, we estimate an increase by between 1.9% and 3.8%, although the estimates are statistically significant only in the case of the DID-All sample. For the RDD, the effects range from 17.7% to 24.7%

and are statistically significant for the RDD-IK and RDD-None samples.

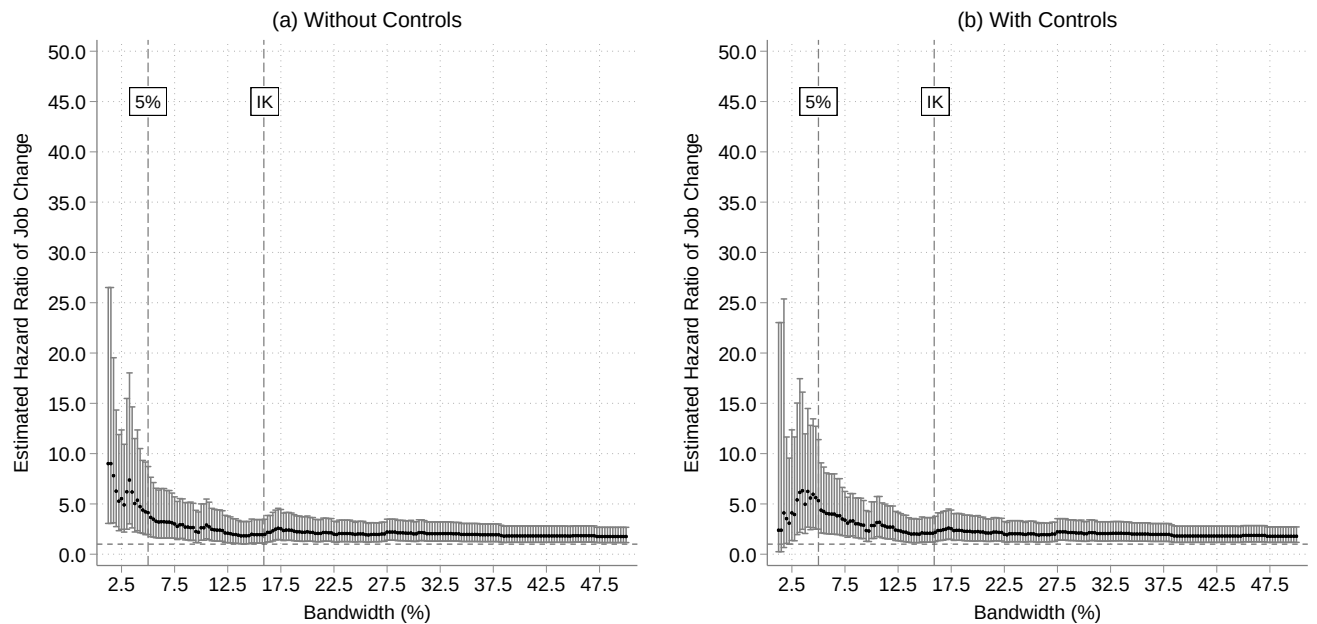
These effects should be interpreted with caution because the underlying data are subject to measurement error, given the often imputed and coarse salary values. That said, the measurement error involved pertains to the dependent variable. As such, as long as the error is not systematically associated with election to moderatorship, the error merely implies a lack of precision in our estimations, rather than bias (Hausman 2001). Consistent with this idea, we note that the estimates are consistently positive, with some turning insignificant only upon adjustment for imputation (Online Appendix Tables A.2 and A.3).

6. Robustness

6.1. Sensitivity

The Online Appendix reports several sensitivity tests, side-by-side, for both the DID and the RDD. The

Figure 4. Regression Discontinuity Estimates for Various Bandwidths



Notes. This figure plots the RDD hazard ratio estimates for bandwidths between 1% and 50% (in 0.25 percentage point steps). Panel (a) is Equation (2) without controls. Panel (b) is Equation (2) with controls. *IK* denotes the bandwidth of Imbens and Kalyanaraman (2012) used for RDD-*IK*. 5% marks the 5%-bandwidth used for RDD-5%. The dashed horizontal line denotes a hazard ratio of 1. Whiskers give 90% confidence intervals. Black dots mark the point estimate.

results for job changes are consistent when using a Logit estimator (Online Appendix Tables A.4 and A.5). The RDD results for job changes and salary are consistent, using pooled, quadratic, and nonparametric specifications (Online Appendix Tables A.6–A.9). Regarding job changes after elections, 8.39% are within-employer and 0.19% reflect employment by Stack Overflow, Inc. Therefore, we also compare the estimates when omitting job changes from the measure that involve Stack Overflow, Inc. as the employer, the company operating the communities, or omitting company-internal job changes (i.e., by recoding them as no job change; Online Appendix Table A.10). Finally, we test the sensitivity toward

the exclusion of subsamples. First, we explore dropping candidates from the Stack Overflow community, the largest community on Stack Exchange. Dropping candidates reduces the sample size and therefore statistical power; thus, to enable a careful interpretation, we proceed stepwise, dropping 10% of the candidates at random and reporting the estimate, then 20%, and so on. Overall, the results are robust to removing these candidates, with the exception that for DID-All the effects fall below the significance level when we omit 80% and more of the candidates (Online Appendix Figure A.1). Second, we explore dropping repeat candidates (i.e., candidates who ran for election on multiple occasions)

Table 6. Comparison of Effect Magnitudes

	Hazard ratio	Predicted probability		
		Winner	Nonwinner	Difference
DID-CEM	1.653	29.6%	21.9%	7.8
DID-PSM	1.895	31.5%	21.1%	10.4
DID-All	1.233	29.4%	24.7%	4.7
RDD- <i>IK</i>	1.979	43.5%	31.6%	11.9
RDD- <i>IK</i> , with controls	2.174	43.2%	30.8%	12.3
RDD-5%, with controls	5.328	44.4%	34.7%	9.7
RDD-None, with controls	1.789	39.7%	29.3%	10.4

Notes. This table reports the average predicted probabilities of a job change. Column “Hazard Ratio” reports the estimates based on Tables 4 and 5. The last column reports the difference in means between the probabilities. Predicted probability for winners and nonwinners is rounded to the first digit. Column “Difference” is given in percentage points and computed based on the exact (unrounded) probabilities for winners and nonwinners.

Table 7. Difference-in-Differences Estimates of Salary Effects

	DV = Log(Salary)		
	(1)	(2)	(3)
<i>After x Mod</i>	0.036 (0.032)	0.019 (0.025)	0.038* (0.022)
Sample	DID-CEM	DID-PSM	DID-All
N (candidates/observations)	157/760	309/1,515	472/2,281

Notes. OLS estimates. Observations are candidate-half-years. The number of observations is smaller than in the main models because the dependent variable is not observed for all candidates. Standard errors are in parentheses. Coefficients and standard errors are rounded to three decimal places.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

in a similar manner. The findings are consistent; the only exception is that when 90% and more of the repeat candidates are dropped, RDD-IK drops below the significance level (Online Appendix Figure A.2).

6.2. Checks for Model Assumptions

Regarding the DID, the Online Appendix reports tests for parallel trends (Angrist and Pischke 2009). We interact *Mod* with time dummies for the chronological distance from the election. None of the pre-election interactions is significant, and thus, there is no indication that the parallel trends assumption is violated (Online Appendix Table A.11). After the election, we observe significant interactions, which confirms the main results. To further ensure that the estimate is not an artifact of pre-existing differences between winners and nonwinners, we implement a placebo test (Bertrand et al. 2004). The test seeks to assess whether the effect arises in data where the null is known to hold. We follow prior work (Brynjolfsson et al. 2019, Shin et al. 2023, Chan et al. 2024) and restrict the estimation to the pre-election period and then create an artificial treatment event one year before the actual election

(i.e., we move the treatment to the middle of the pre-period). We find no statistically significant effect of the placebo event, which corroborates the assumptions (Online Appendix Table A.12). In summary, we find no evidence that the parallel trends assumption is violated.

Regarding the Cox model, we assess the proportional hazards assumption (Cox 1972, Lancaster 1990), that is, that the hazard ratio remains constant over time. Figure 5(a), provides the standard log-log plot for the full sample of candidates. The curves are reasonably parallel, which supports the proportional hazards assumption. We implement a test of the relationship between scaled Schoenfeld residuals and time for all subsamples (Grambsch and Therneau 1994). None of the tests indicates a violation of the assumption.

For the RDD, we test the no sorting assumption, meaning that candidates are not able to self-select or self-sort into winning the election, for example, by manipulating the election outcome. We implement the standard test in its conventional and bias-corrected version (McCrary 2008, Calonico et al. 2014, Cattaneo et al. 2018). The test seeks to reject the null of clumping in the kernel density on either side of the vote share

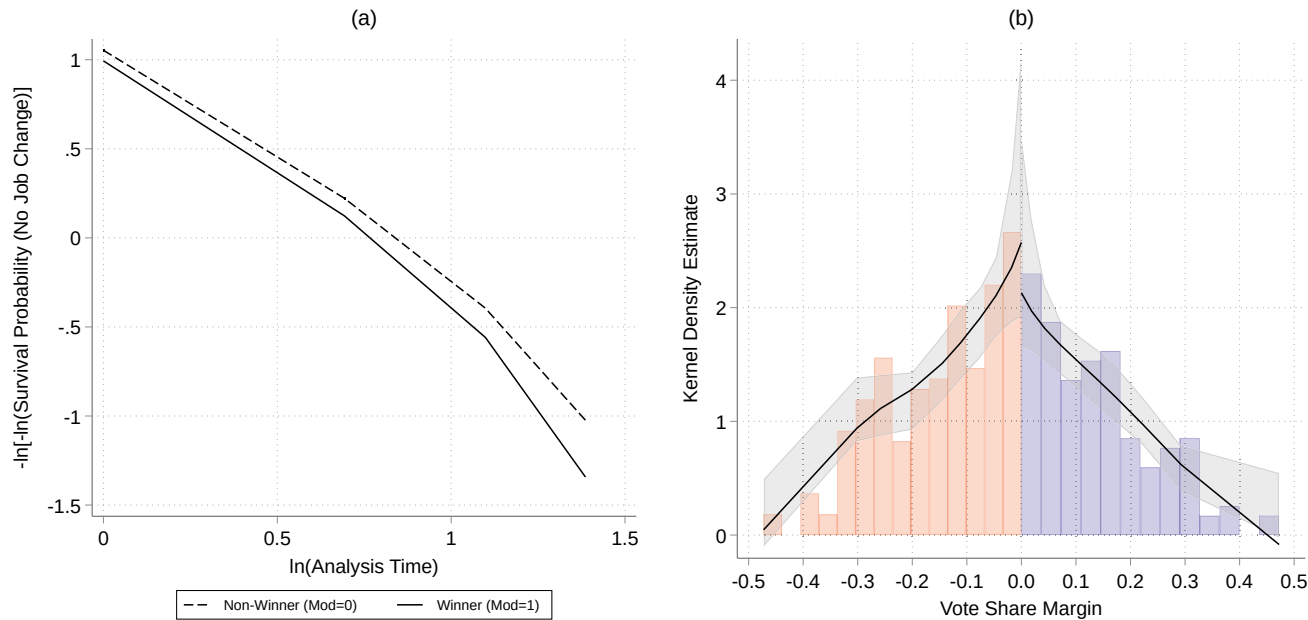
Table 8. Regression Discontinuity Estimates of Salary Effects

	DV = Log(Salary)			
	(1)	(2)	(3)	(4)
<i>Mod</i>	0.247** (0.113)	0.238** (0.114)	0.242 (0.169)	0.177** (0.080)
<i>Share margin</i>	−0.435 (1.103)	0.510 (1.218)	−5.029 (4.611)	0.163 (0.356)
<i>Mod x Share margin</i>	−0.959 (1.490)	−2.234 (1.651)	10.409* (6.062)	−0.779 (0.510)
Controls		x	x	x
Sample	RDD-IK	RDD-IK	RDD-5%	RDD-None
N (candidates)	175	175	70	286

Notes. OLS estimates. *N* gives the effective sample size in candidates. *N* is smaller than in the main models because the dependent variable is not observed for all candidates. Standard errors are in parentheses. Coefficients and standard errors are rounded to three decimal places.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

Figure 5. (Color online) Graphical Assessment of Model Assumptions



Notes. Panel (a) plots the log-log Kaplan Meier survival estimates against the log of time based on the full sample of candidates. Solid curve is for winners, dashed curve for nonwinners. Panel (b) plots the kernel density of the vote share margin along with 90% confidence intervals based on the procedure in Cattaneo et al. (2018). Density plot is based on a triangular kernel.

threshold. The tests do not lend evidence to a violation of the assumption ($p_{\text{conventional}} = 0.41$, $p_{\text{robust}} = 0.52$). Figure 5(b), plots the kernel density distribution of the vote share margins and 90% confidence intervals. There is no evidence of a discontinuity at the cutoff, which supports the assumption.

6.3. Alienation of Nonwinners

The positive effects we observe could be due to negative responses of those who fail to win an election. Were this to happen, it would violate the stable unit treatment value assumption, or SUTVA (Cox 1958, p. 19), and invalidate the use of nonwinners as comparison benchmark. To assess, we use the panel data samples (i.e., DID-CEM, DID-PSM, DID-All) restricted to non-winners and estimate a simple pre-post regression model of job changes, also including a control for a general time trend. The resulting coefficients are insignificant, which refutes concerns over alienation (Online Appendix Table A.13).

6.4. Validity of Applying the RDD in an STV Setting

One could be concerned that the STV's round-by-round reallocation of votes artificially depresses voter preferences, as indicated by the first preferences. If so, the distances between the most-voted nonwinner and least-voted winner might appear smaller than the disparity in voters' true preferences. This should not be

concerning for several reasons. First, although the STV procedure is not without criticism, we are not aware of any such bias being reported in prior studies (Hill 1988, Farrell 2011). Instead, the STV procedure helps to eliminate the hopeless nonwinners, and identifies clear winners, up front. The STV procedure then reallocates the remaining candidates to seats depending on expressed voter preferences while eliminating those candidates with relatively less preference from voters. Online Appendix Figure A.3 corroborates this fact in our data: comparing the distribution of first preferences, we see that the STV procedure eliminates the hopeless nonwinners at the left tail of the first-preference distribution as well as the clear winners at the right tail. Further corroborating is that our regression results hold when matching candidates based solely on their share of first preferences (Online Appendix Tables A.14 and A.15).

6.5. Sampling Bias

Our study design depends on the assumption that the availability of LinkedIn profile data is not systematically associated with whether an individual wins or loses an election. Furthermore, it assumes that censoring in a candidate's job change time series must be independent of treatment assignment (see, e.g., Bor et al. 2014). One concern could be that the election outcome influences whether a candidate opens a LinkedIn account. Refuting this concern, we find that there is no statistically significant difference between winners and

nonwinners on LinkedIn data availability, also not when conditioning on election year or community theme (Online Appendix Table A.16). A related concern is that the election outcome encourages candidates to put more effort into self-presentation on LinkedIn. Our data do not indicate when a profile was updated, such that we cannot directly test whether winning an election is associated with more information being reported in the profiles. An ex post comparison of winners' and nonwinners' LinkedIn profiles along various characteristics, including profile description length, endorsements received, and accomplishments, does not indicate any statistically significant difference (Online Appendix Tables A.17 and A.18).¹¹ Thus, at least from an ex post angle, winners and nonwinners do not differ along several pieces of nonjob-change-related information, which provides some confidence that the findings are not driven by changes in reporting behaviors.

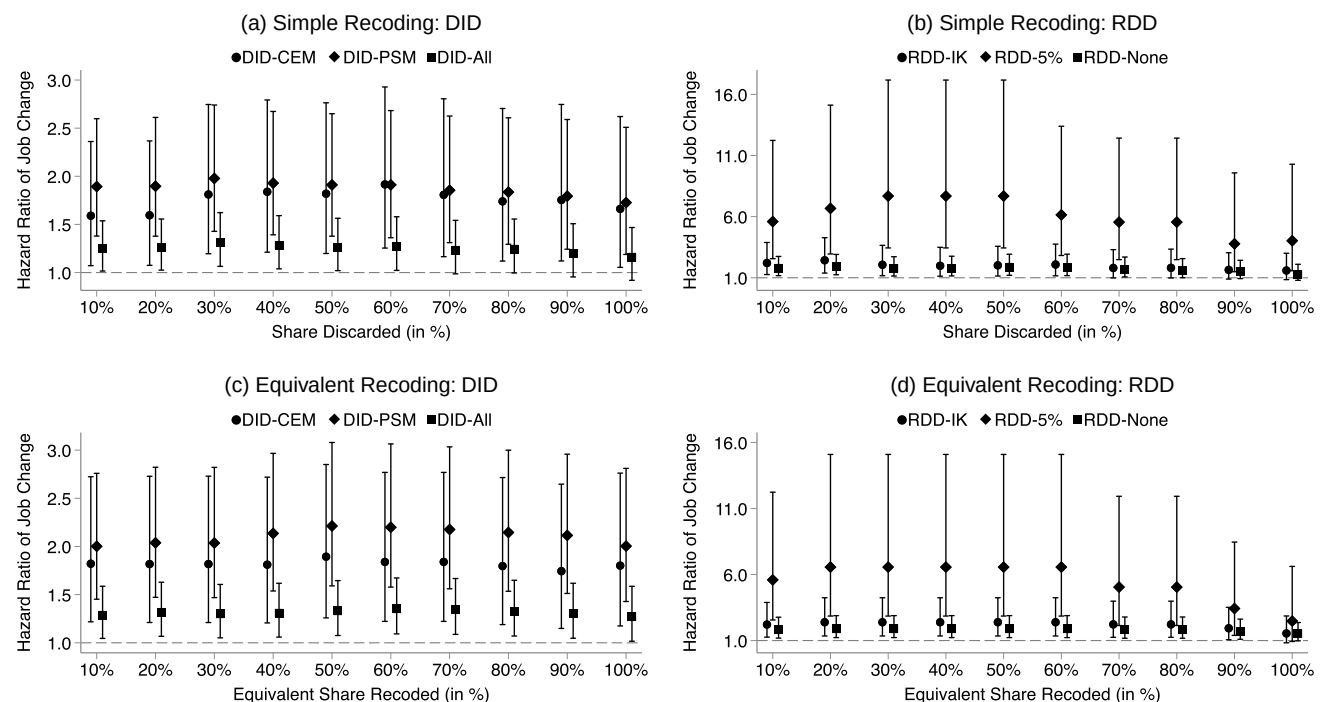
6.6. Employment Gaps

It is important to consider job changes with and without employment gaps because job changes with employment gaps could indicate a period of unemployment (i.e., a career harm rather than benefit). For our job change measure, we had chosen to retain jobs with employment gaps because such gaps are common and

may be driven by a variety of factors, such as taking time off to raise a family, pursuing education or training, traveling, or taking a break from work for personal or professional reasons; that is, they will frequently not be due to layoffs. Nevertheless, we explore the stability of our results to an adjusted measure that excludes job changes with employment gaps. The goal of this exercise is to ensure that those job changes are not driving our findings.¹² The key challenge for such a test is that discarding observations artificially reduces variation in the outcome, so any loss of significance might be due to a decline in statistical power. Therefore, we proceed stepwise, as we did in Section 6.1; we begin by recoding 10% of the job changes with employment gaps (chosen at random) as 0s (i.e., as no changes) and plot the resulting estimate, then 20%, and so on.

Figure 6, (a) and (b), report the estimates for Equations (1) and (2). The estimates from the DID-CEM, DID-PSM, and RDD-5% estimation remain significant even when we recode all job changes with employment gaps. For DID-All and RDD-IK, the estimates do not lose significance until we recode 70% and more of the relevant job changes. The RDD-None estimates are significant until we recode 90% and more. Figure 6, (c) and (d), report a complementary test in which we recode the equivalent number of job changes without

Figure 6. Robustness Check on Employment Gaps



Notes. Panels (a) and (b) plot the hazard ratios obtained from Equations (1) and (2), depending on how many of the job changes with employment gaps are recoded as no job change at random, for the DID and RDD subsamples. Panels (c) and (d) plot the hazard ratios obtained from Equations (1) and (2) when the equivalent number of job changes with employment gaps are recoded, at random, as 'no job change' in the DiD and RDD subsamples. The markers give the point estimate, and the whiskers are 90% confidence intervals.

employment gap at random. For example, if candidate A experienced a job change with an employment gap, we recode another randomly selected candidate B's job change—which did not show an employment gap—a 0. Here, the RDD estimates turn insignificant when 100% is recoded. These observations support that job changes with employment gaps are not fundamentally driving the results; rather, omitting them is likely leading to a loss of statistical power.¹³

7. Mechanism Exploration

Admittedly, identifying the causal mechanisms that underlie the treatment effect poses a greater challenge than estimating the treatment effect itself. In the following, we present exploratory results, which are at best suggestive of the mechanisms at work.

7.1. Human and Social Capital

To explore the human and social capital mechanisms, we use the indirect effects estimation of Buis (2010), Hicks and Tingley (2011), and Imai et al. (2011). The analysis seeks to isolate the portion of the causal effect of moderatorship on job changes that is attributable to improvements in human or social capital. This requires the estimation of two equations:

$$M_i = \alpha_1 + \beta_1 \text{Mod}_i + \epsilon_i, \quad (3)$$

$$\text{Logit}(\text{Job change}_i) = \alpha_2 + \lambda_1 \text{Mod}_i + \lambda_2 M_i \quad (4)$$

where M denotes the mediator (i.e., *Tag badges* for human capital and *Chat contacts* for social capital), and the remainder of the variables and indices follow the same notation as in Equation (2). Equation (3) is estimated with OLS; Equation (4) is estimated with Logit. For a mediation to be present, three conditions need to be met. First, in Equation (3), Mod has a positive effect on M ($\beta_1 > 0$). Second, in Equation (4), M has a positive effect on *Job change*, even when controlling for Mod (i.e., $\lambda_2 > 0$). Third, the natural indirect effect (NIE) is positive. The NIE is the estimate of the effect that Mod exerted on *Job change* through M . The mechanics for calculating the NIE is to use Equation (3) to simulate predictions for M in both the $\text{Mod} = 0$ and $\text{Mod} = 1$ case, to subsequently plug these predicted M into Equation (4) for $\text{Mod} = 0$ and $\text{Mod} = 1$ to simulate predictions on *Job change*, and finally compare the average differences in outcomes for both cases. The NIE is expressed in original units or as a share of the total effect.¹⁴

The mediation test depends on the assumption of sequential ignorability, meaning that M is independent conditional on Mod . In nonlinear models, the standard difference of coefficients methods for estimating the average causal mediation effect (ACME) will no longer provide a consistent estimate under sequential ignorability (VanderWeele 2009, Imai et al. 2011). For these

reasons, we use the procedure of Hicks and Tingley (2011), who extended the indirect effects approach to nonlinear models. This procedure first fits the models for outcome and mediator variables, then simulates the model parameters from their sampling distribution, then afterward simulates potential values for the outcome and the mediator for each draw of model parameters, and then computes the statistics of interest. Online Appendix B provides the technical details.

Table 9 shows the results. We cannot confirm a mediation effect for human capital; none of the models supports a significant effect of *Mod* on *Tag Badges* (i.e., columns 1, 3, and 5). We observe evidence for the social capital mechanism. All three conditions for mediation are met; *Mod* has a positive effect on *Chat contacts* (i.e., columns 7, 9, and 11) yet is not significant for RDD-5%; there is a positive association between *Chat contacts* and *Job change* (i.e., columns 8, 10, and 12); and the NIE is positive.¹⁵

Regarding the sensitivity, Online Appendix Figure B.2 plots the results of the ρ -sensitivity analysis (Hicks and Tingley 2011). The analysis quantifies the degree of sensitivity by looking at how large the parameter ρ , the correlation between the error terms, must be for the ACME to be insignificant. Generally, the larger ρ , the more robust the results are toward unobserved confounders. The results show that, to conclude that the true ACME for RDD-IK is not statistically different from 0, we must assume that an unobserved confounder that affects both social capital and job change in the same direction and makes the correlation between the error terms greater than 0.30. There are no objective standards for assessing the absolute values of ρ , but it indicates a relative level of robustness specific to the empirical setting and model, thus necessitating cross-study comparison (Keele et al. 2015). We are not aware of a study directly comparable to ours; however, Dimitriadis and Koning (2022) studied the effects of a social skills training program on entrepreneurs' profits (mediated by social interactions among them) and reported $\rho = 0.16$. Carpena and Zia (2020) reported the findings of a field experiment on the returns to financial education and considered a $\rho = 0.23$ as supportive of the mediation. Our ρ is larger. Furthermore, we can consider the importance of the unobserved confounder in explaining the observed variation. For RDD-IK ACME to be 0, for example, an omitted confounder must explain 30% of the remaining variance in the mediator and 30% of the remaining variance in the outcome (i.e., $R_M^2 \times R_Y^2 = 0.09$ must be satisfied).

There is also anecdotal evidence for the social capital mechanism. On Stack Exchange, we found several posts of moderators in which they described the social component of their role. Part of this involves interacting and bonding with other moderators, for example, "A good amount of my day is spent hanging out in the

Table 9. Mediation Tests for Human and Social Capital

	Y = <i>Job change</i> , M = <i>Log(Tag badges)</i>				Y = <i>Job change</i> , M = <i>Log(Chat contacts)</i>							
	DV = M (1)	DV = Y (2)	DV = M (3)	DV = Y (4)	DV = M (5)	DV = Y (6)	DV = M (7)	DV = Y (8)	DV = M (9)	DV = Y (10)	DV = M (11)	DV = Y (12)
<i>Mod</i>	0.151 (0.296)	2.461* (1.345)	0.165 (0.446)	7.348** (7.141)	0.198 (0.212)	2.275** (0.877)	0.179* (0.091)	2.149 (1.182)	0.073 (0.152)	7.306** (6.947)	0.239*** (0.057)	1.818 (0.719)
<i>Share margin</i>	0.503 (2.487)	78.801 (361.036)	-2.394 (11.468)	0.043 (0.977)	0.824 (0.756)	11.193* (16.114)	0.407 (0.766)	47.978 (219.308)	1.993 (3.900)	0.001 (0.012)	-0.112 (0.202)	13.946* (20.248)
<i>Mod x Share margin</i>	2.796 (3.589)	0.000 (0.001)	6.607 (16.677)	0.000 (0.000)	0.491 (1.168)	0.005** (0.011)	-0.091 (1.105)	0.000 (0.002)	2.523 (5.671)	0.000 (0.000)	-0.517* (0.312)	0.007** (0.016)
M		1.377** (0.200)		2.170** (0.660)		1.038 (0.104)		4.424** (2.774)		6.504* (7.079)		3.512*** (1.643)
NIE	0.011 [-0.023, 0.050]		0.019 [-0.062, 0.106]		0.002 [-0.008, 0.014]		0.057 [0.007, 0.121]		0.020 [-0.047, 0.100]		0.069 [0.026, 0.117]	
%TE mediated	0.048 [0.023, 0.216]		0.056 [0.037, 0.179]		0.009 [0.005, 0.029]		0.241 [0.126, 0.884]		0.059 [0.039, 0.174]		0.322 [0.189, 0.925]	
Sample	RDD-IK	RDD-IK	RDD-5%	RDD-5%	RDD-None	RDD-None	RDD-IK	RDD-IK	RDD-5%	RDD-5%	RDD-None	RDD-None
N (candidates)	187	187	77	77	323	323	187	187	77	77	323	323
Estimation	OLS	Logit	OLS	Logit	OLS	Logit	OLS	Logit	OLS	Logit	OLS	Logit

Notes. Logit coefficients reported as odds ratios. N gives the effective sample size in candidates. Standard errors are in parentheses. Coefficients, standard errors, and effect sizes rounded to three decimal places. Based on the output of the medeff package, we report NIE and %TE mediated together with 90% confidence intervals. NIE, natural indirect effect; TE, total effect. *, **, *** indicate significance at the 10%, 5%, and 1% levels, respectively.

private chatroom for all of the Stack Exchange moderators. We talk to other moderators about moderation issues [...] Additionally, moderators mingle with power users, that is, users who are strongly engaged and support moderators. There are chatrooms on Stack Exchange in which power users of a community can exchange information with each other and the moderators.

7.2. Signaling

To explore signaling, we implemented the test formally modeled by Graetz (2017, 2021). It is based on the idea that signaling effects should deteriorate over time because of employers’ learning about the true quality of a candidate. If signaling is at play, we should observe that career benefits manifest relatively soon after the election. By contrast, human capital and social capital take time to accumulate, and thus effects resulting from these mechanisms should manifest after some time in the position. Therefore, we estimate the RDD separately for each half-year after the election, as reported in Table 10. We observe significant effects in t and $t + 2$. Interpreting these results via the model proposed by Graetz (2017, 2021), the immediacy of the observed effect in t is consistent with a signaling mechanism. The relative-time DID model corroborates this observation (Online Appendix Table A.11), as do the diverging curves in Figure 2.¹⁶

8. Discussion

Our study makes two main contributions. First, we contribute to the debate on career benefits from participating in online communities (Lerner and Tirole 2002, Roberts et al. 2006, Hann et al. 2013). Departing from Lerner and Tirole (2002), several studies have suggested that community participation may be driven by an expectation that it leads to employment benefits (Hann et al. 2013, Huang and Zhang 2016, Xu et al. 2020). Causal evidence of community engagement leading to career benefits has so far been lacking. Our study provides such evidence, particularly focusing on community leadership. Demonstrating that contributors’ expectations are well-founded is important because, were this not the case, the resultant failure to manage contributor expectations would likely have undesirable long-run consequences for community commitment. Had we found little evidence of career benefits the natural implication would be for the platform operator to highlight other benefits of participation when attracting and engaging with members, for example, community or a sense of belonging. Given that we have confirmed the validity of career benefits, the implication is that operators of communities should indeed highlight those benefits to members.

Table 10. Tests for Signaling Mechanisms

	DV = Job change							
	(1) <i>t</i>	(2) <i>t</i> + 1	(3) <i>t</i> + 2	(4) <i>t</i> + 3	(1) <i>t</i>	(2) <i>t</i> + 1	(3) <i>t</i> + 2	(4) <i>t</i> + 3
<i>Mod</i>	5.659** (3.897)	0.626 (0.438)	2.034 (1.729)	0.799 (0.601)	7.114* (7.339)	3.600 (3.618)	1.009 (1.222)	0.418 (0.539)
<i>Share margin</i>	0.005 (0.029)	345.262 (2,035.998)	0.003 (0.025)	2,892.833 (23,435.906)	1.140e + 15** (1.996e + 16)	0.000 (0.000)	1.505e + 66* (1.218e + 68)	0.000 (0.000)
<i>Mod × Share margin</i>	0.081 (0.661)	0.001 (0.011)	7.113e + 6* (6.635e + 7)	0.000 (0.003)	0.000* (0.000)	0.000 (0.000)	0.000 (0.000)	1.041e + 25 (4.396e + 26)
<i>Sample</i>	RDD-IK	RDD-IK	RDD-IK	RDD-IK	RDD-5%	RDD-5%	RDD-5%	RDD-5%
<i>N (Candidates)</i>	187	187	187	187	77	77	77	77

Notes. Logit estimation. Coefficients reported as odds ratios. *N* gives the effective sample size in candidates. Standard errors are in parentheses. Coefficients and standard errors are rounded to three decimal places.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

Second, we also contribute to research on the motivations behind private contributions to online communities (see, e.g., Wasko and Faraj 2005, von Krogh et al. 2012, Wang et al. 2022). The notion that economic motives drive contributions has long been advocated, and relatively strong evidence exists that the perception of career benefits drives individuals' contributions. However, our study presents the first rigorous empirical evidence that this perception is a reality.

Finally, our study provides exploratory empirical evidence about possible mechanisms underlying resulting career benefits. That we find support for a social capital explanation complements prior explanations of human capital benefits as a driver, although we also, notably, find no evidence in support of the latter. What remains unclear, however, is the extent to which social capital benefits are perceived by contributors, especially compared with signaling benefits or human capital development. Understanding contributor expectations about the role of specific mechanisms is important. If individuals purposefully engage in communities with incorrect expectations about their signaling or the degree to which that engagement will result in the acquisition of market-relevant skills, then they are likely to engage in a suboptimal manner, selecting into the *wrong* activities, for example. Future work can look to address this aspect.

At the same time, understanding participant perceptions about specific mechanisms behind possible career benefits is also important for the design and management of online communities. For example, if community participants are not aware of the important role of social capital development as an underlying mechanism for career benefits, then they may not be sufficiently engaged in forming new social ties or engaging with new members, etc. Conversely, if participants believe that social capital benefits are a useful return to their engagement, then policies and mechanisms employed within communities are best designed with a focus on facilitating social engagement, and less so on confirming expertise, or on skill acquisition.

It would also be valuable to assess the external validity of our findings with respect to other types of communities, for example, those focused on open-source software or online encyclopedias. Notably, Wikipedia also holds elections for leadership positions, which could be exploited to apply a similar econometric framework. One limitation of our research is that we cannot perfectly observe career benefits but have to rely on job changes. Another limitation relates to the matching of the LinkedIn profiles. Although our checks did not indicate a sampling bias, future work might consider using alternative data sources to elicit careers, for example, surveys and interviews, as in Hann et al. (2013). Last, it is generally challenging to measure human and social capital and particularly so in the case

of observational data from online sources. Our measures provided the best feasible option. However, future research should consider implementing more robust measures. For human capital, researchers could obtain data on test outcomes (e.g., university exams, ratings of managers, assessment center tests) or schooling (e.g., training hours, certificates). For social capital, researchers could obtain data on actual interactions (e.g., frequency of exchange between individuals, number of meetings attended together). Although we find support for social capital mechanisms using incoming chat contacts, we are unable to examine a measure of reciprocal chat contacts because there is too little variation in the measure. Moreover, our understanding of social capital would be improved here if we could assess which contacts in particular were driving the effects (e.g., power users, other moderators). Regarding the signaling mechanism, we observe some support, but we also observe that the results for later time periods are not consistently significant.

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Endnotes

¹ The expectation is typically that such engagement can improve an individual's human or social capital and serve as a signal of ability to employers. For example, community leaders are expected to improve "connections and networking," "exposure to new technologies," and the "leadership skills" (CIO 2013).

² An additional benefit of focusing on job changes is that this enables comparisons between our findings and those of prior studies that have also considered the relationship between labor and online community contributions (Huang and Zhang 2016, Xu et al. 2020). To validate our claim that positive effects on our job change measure will reflect career benefits, we also consider alternative measures, namely estimates of workers' salaries.

³ For example, see <https://math.stackexchange.com/election>.

⁴ We thank the Associate Editor and an anonymous reviewer for making the observation that tag badges received immediately after an election reflect pre-election effort human capital acquisition on the part of candidates; hence, our measures focus on human capital accrued in a period that lags the election by one half-year.

⁵ We thank an anonymous reviewer for suggesting that we count only incoming messages.

⁶ We thank the Associate Editor for suggesting the use of a hazard framework.

⁷ Thus, there are no *forbidden* comparisons entering the DID estimate that may occur in staggered designs (Goodman-Bacon 2021). In addition, the control observations within elections consist almost entirely of never-treated candidates (91%).

⁸ We thank the review team for suggesting the matching variables.

⁹ The RDD yields an estimate of the LATE, in theory mitigating the need for an extensive set of controls (Imbens and Lemieux 2008). Still, adjusting our estimates for heterogeneity can yield more precise estimates. This gain in precision comes at the expense of estimating additional parameters. This is a challenge because the sample size within the bandwidth is relatively lower, implying relatively limited statistical power. Moreover, our RDD is cross-sectional and therefore cannot account for candidate fixed effects.

¹⁰ We thank the review team for suggesting this discussion.

¹¹ We thank an anonymous reviewer for suggesting this test.

¹² We thank the Associate Editor for suggesting these analyses.

¹³ Inspired by Xu et al. (2020), we also assess candidates' participation following a job change and observe declines in several participation outcomes, as expected. We thank an anonymous reviewer for this idea (Online Appendix Table A.19).

¹⁴ A hazard estimation in this case is not feasible because mediation models are still under development for such estimators.

¹⁵ Online Appendix Table A.20 provides the results when employing an alternative measure of social capital, namely outgoing chat messages.

¹⁶ It is important to be cautious regarding the disappearance of statistical significance of the treatment effect in some periods of Table 10. The result in these cases classifies as a null result and as such does not necessarily indicate the absence of an effect or relationship. These estimates could be insignificant because they are noisier. Online Appendix Table A.11 uses the larger DID sample, and there we observe significant effects in earlier as well as later post-periods. A second explanation for this is that quality signals in the job market tend to have stronger effects in the short run, rather than driving stable and consistent effects over the long term (Connelly et al. 2011, Khoo and Ost 2018). Once a signal has been obtained, a candidate can benefit immediately.

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